A Data-Driven Exploration of Movie Success

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Abstract

These days, ratings on movie platforms play a crucial role in shaping audience perceptions and the industry success on a movie but we should consider more than just a few scores.This study explores the factors contributing to finding the trend of successful movies by leveraging data from three major platforms: IMDb, TMDb, and Rotten Tomatoes. After going through TMDb's top 300 movies through their API and collecting additional information by web-scraping IMDb and Rotten Tomatoes, we found that the trends of ratings have a slightly moderate positive relationship between all three platforms along with with many other factors that can help define success in the film industry

Introduction

Movies have long been a cornerstone of entertainment, offering not just enjoyment but a sense of connection and shared experience. Whether it's the magic of watching a film for the first time in a darkened theater or, in our modern day world, the convenience of streaming on a laptop in the comfort of your own home, movies have always had the power to captivate and engage audiences. However, beyond the emotional and cultural impact of films, there is a growing need to understand the trends and factors contributing to a movie's success in the modern competitive industry.

For many, the ratings of a movie from various popular platforms are often seen as the primary and ultimate factor in determining its success. However, it can be far more complex. A movie's popularity can be influenced by many factors ranging from audience reception to production budgets. Understanding the interplay between these elements is essential for

gauging what makes a film successful in the ever-evolving entertainment landscape. With the advancements of data-driven tools, we have the opportunity to explore the factors and approach our questions in a systematic, quantifiable way.

Motivation and Data Acquisition

In this study, we decided to look into three of the most popular movie review platforms on the internet which were the Internet Movie Database (IMDb), The Movie Database (TMDB), and Rotten Tomatoes. These platforms allowed us to capture a variety of factors, including audience ratings, critical scores, and popularity metrics, ensuring a near comprehensive approach to analyzing movie performance from each of these websites influences. Our goal is to understand how these ratings and factors intersect, and how other movie details can provide a broader understanding of success in today's competitive entertainment landscape.

Our approach was to collect the top movies from one of these websites and while IMDb and Rotten Tomatoes have the historical credibility, we ultimately chose the top 300 movies from TMDb that bodied a globally representation of the modern audience in movies with diverse cultures. Using TMDb's API, we retrieved data on the top 300 rated movies, which formed the basis of our analysis. By including movies from multiple regions and cultures, we ensured that our dataset represented a broad spectrum of movie success, making it more relevant for understanding trends in the current film industry.

While TMDb provided a robust dataset, we were still adamant about incorporating detailed information from IMDb and Rotten Tomatoes. We decided to web-scrape these two websites for the relevant data to our studies and this allowed us to get the obtainable features and aspects of each platform. From IMDb, we were able to get other information such as budget, gross worldwide, runtime (in minutes), country/countries of origin, and release date. Most of this was part of a section called Details within all IMDb pages. Due to some restrictions

in web scraping IMDb, we turned to Rotten Tomatoes for the remaining information we desired, including the director, production company, movie age rating, and genres.

While we were unable to recover all 300 movies from TMDb across all three platforms, we were able to collect data for 278 movies which we believe is a large enough dataset for meaningful analysis. Ultimately, this approach allows us to draw information from a diverse set of data sources, and allows us to explore the necessary factors that influence a movie's success. By combining ratings, and production details, we hope to uncover insight into what makes a movie successful giving a new and improved data analytics perspective into making new movies in the future.

Explaining challenges

One of the primary challenges was working with TMDb's API. While TMDb provides some documentation and examples for certain endpoints in its API reference, identifying the correct approach to retrieve the Top 300 movies required some exploration and experimentation. Much of TMDb's API reference focuses on user-specific information, such as retrieving top-rated movies or watchlists based on an account_id endpoint. After examination, we found the right endpoint to be "discover" which allows users to query movies based on various criteria, including release dates, vote averages, and other filters. Two of the filters used were essential to our results which were vote_count.gte and page. While sort_by was the main filter that allowed us to sort the top rated movies in TMDb, this resulted in many undesirable movies with high ratings but very few reviews. The vote_count filter ensures that the movies returned by the API have been rated by at least 300 users making our data more robust and relevant for identifying trends in movie success. Additionally, handling pagination by running the page parameter through a for loop was necessary to retrieve all 300 movies, as the API returned movies in batches of 20 movies per page. Combining these filters and parameters allowed us to refine our movie dataset and focus on the real top rated movies.

Another challenge we faced was looking up specific movies on IMDb. Unlike some platforms that use a predictable or structured URL format for each movie, IMDb's URLs are unique and include an alphanumeric identifier (e.g. tt0111161 for The Shawshank Redemption). This pattern made the search difficult instead of simply plugging in the movie titles at the end of an IMDb url resulting in a redirect to the movie rating page. To overcome this, we had to rely on IMDb's advanced search page, which provides a feature for inputting movie titles. This allowed us to retrieve a list of related movies, from which we matched titles to extract the correct IMDb links. While mostly effective by giving back over 90% of the movies from TMDB, this process added an extra step to streamline data collection and highlighted a need in the future for a more direct access method to collect certain IMDb movie pages.

A question you may ask however, is how successful was it to connect the titles within all three movie websites. Comparing titles between TMDb and IMDb was mostly straightforward, as the titles were formatted similarly on both websites. However, the same cannot be said for the transition into Rotten Tomatoes movies. Rotten Tomatoes often uses alternate titles or slight variations in formatting which makes it more difficult to establish direct matches. This discrepancy required additional effort by making a supplementary function called strip_punctuation which creates a transition table that removes all of the punctuation from any given string. For example, the movie title, “Spider-Man: Into the Spider-Verse” was reduced down into, “SpiderMan Into the SpiderVerse”. Once the function was applied, this solution resulted in a significant improvement, increasing the total number of matched movies in our dataset from 238 to 278. This highlights the importance of preprocessing and cleaning data to overcome inconsistencies of movie titles across all three platforms.

Our final challenge web-scraping details within each Rotten Tomatoes movie page once the connection was made between all three databases. Information such as the Tomatometer (rt_score) and Ppcornmeter (pop_score) were easy to scrap but there was some difficulty with the Movie Info section. The challenge stemmed from the fact that every line of information in the

"Movie Info" section, ranging from the synopsis to the aspect ratio, shared the same HTML tag structure. This made it difficult to isolate and extract specific details through soup's simple 'find' function, as there were no unique identifiers or tags to differentiate between elements. We ended using Beautifulsoup to identify all relevant sections of the "Movie Info" using the category-wrap class and search for all the labels, item_labels, that were the titles of each detail line. Depending on the label, it then used the appropriate logic to extract the associated data. We used the "find_next" method to locate the corresponding text and following that, tags such as "rt-link" and "rt-text" to scrap the information and append their text. This method ensured that the script could reliably extract and organize the data despite the unique HTML structure of the Rotten Tomatoes movie websites.

Preliminary interpretation

We started with a list of key questions to address with this analysis:

(1) What is the relationship between the ratings?

(2) What is the budget trend like over the years?

(3) What were some of the most successful directors and production companies over the years?

(4) What are some informative words in the movies' description? For the top 50 movies by IMDB rating, what are their tags and how are they similar?

To gain an understanding of the relationship between the three different ratings, we first run a 3D linear regression of the ratings as variables. In our model, RT score was used as the dependent variable, and IMDB and TMDB ratings as the independent variables. The resulting coefficient for IMDB rating was about 0.66 and for TMDB was about 0.30, with an intercept at 1.09. This suggests that holding all else equal, a one-point increase in IMDB rating would correlate with a 0.66 points increase in RT score; a one-point increase in TMDB rating would correlate with a 0.30 points increase in RT score. The baseline value, where IMDB and TMDB ratings are 0, for RT was 1.09.

Overall, the independent variables were able to account for approximately 10.76% of the variability in RT score. To further investigate, we generated a correlation and covariance matrix for the ratings.

Table 1

Correlation Matrix	IMDB	TMDB	RT
IMDB	1	–	–
TMDB	0.42	1	–
RT	0.33	0.17	1

Table 2

Covariance Matrix	IMDB	TMDB	RT
IMDB	0.32	–	–
TMDB	0.04	0.02	–
RT	0.22	0.03	1.43

Table 1 indicates that the correlation between IMDB and TMDB is the strongest at 0.42; then followed by that between RT and IMDB at 0.33, and lastly RT and TMDB at 0.17. For RT, it is most strongly related with IMDB, which is validated by the previous 3D regression. For Table 2, the variances of the ratings are given in the diagonal entries, with IMDB having the greatest variability. RT and IMDB also have the highest covariance, indicating that the degree to which they vary together is high. In addition, it should be noted that all values are positive, reaffirming the directionality of the relationship between the trio.

Then, to address (2), we computed the median and mean movie budgets for each decade from 1920s to 2020s and visualized them with a line graph. Figure 1 shows a steady increasing trend for the median budget. However, the mean budget exhibited a drastic increase from the 1980s to 2010s. It is notable that the substantial spike in mean in the 1990s and 2010s, which could be due to exceptionally high budget movies like Titanic (1997, $200 million) or Avengers: Endgame (2019,

$356 million). The sudden decrease in budgets from the 2010s to the start of 2020s could be possibly explained by the onset and spread of COVID-19 across the globe.

To investigate (3), we defined director success by calculating their mean gross revenue and movie count within the sample, then using these criteria to list the top 10 directors throughout the years. Similarly, for production companies, success was defined as having a high mean gross revenue. Table 3 summarizes the results. From this, one can see that a single or only a few movies with massive worldwide gross revenues was able to place its director and production company on top of the list. A word of caution here is that the count numbers are only a few, which may skew the results. This may be remedied by including more samples than the current selection.

Figure 1

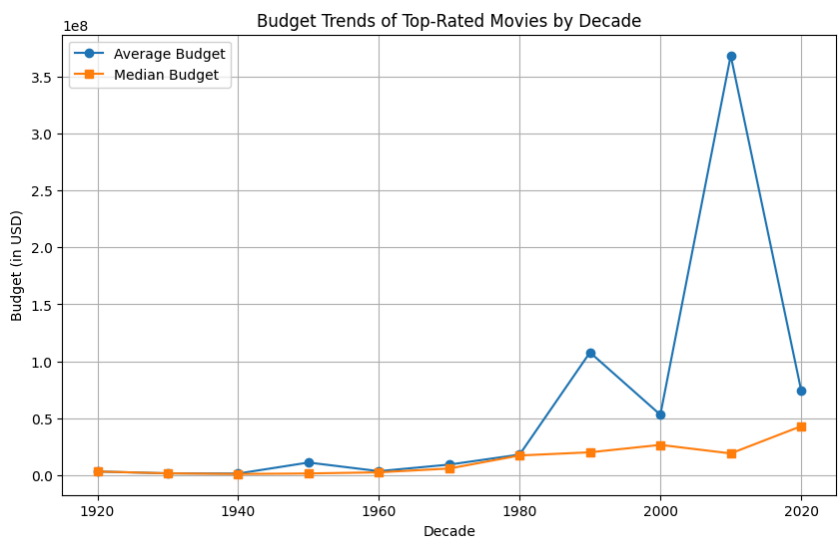

Table 3

Ranking	**Director**	**Mean gross revenue ($1e+08)**	**Count**	**Production company**	**Mean gross revenue ($1e+08)**	**Count**
1	Anthony Russo	24.26	2	Marvel Studios	24.26	2
2	Joseph Kosinski	14.96	1	Don Simpson	14.96	1

3	David Yates	13.43	1	Heyday Films	13.43	1
4	Todd Phillips	10.79	1	BRON Studios	10.79	1
5	Peter Jackson	9.88	3	Fairview Entertainment	9.82	1
6	Jon Favreau	9.82	1	Syncopy	8.60	3
7	Lee Unkrich	8.15	1	Pixar Animation Studios	8.15	1
8	Alfonso Cuaron	8.09	1	New Line Cinema	7.47	4
9	Joapuim Dos Santos	6.91	1	Legendary Pictures	7.14	1
10	Christopher Nolan	6.18	6	Sony Pictures Entertainment	6.91	1

Lasly, to tackle (4), we used the Term Frequency-Inverse Document Frequency method (TF-IDF) to compute the informative values of meaningful words in movie descriptions, or our corpus. The TF portion measures how often a term appears in a text, presenting local importance. The IDF part measures how rare or unique a term is across the corpus; this is designed to designate uninformative words like “the” or “is”. Then TF and IDF are multiplied together to obtain the final value. Using this, we obtained the top 10 most informative words, which are: life, young, world, man, family, war, story, years, time, and new.

We then applied a filter to only obtain the top 50 movies (we do this to reduce the number of resulting clusters and number of movies in each later on) and applied K-means clustering on the TF-IDF values of the movies to generate 5 clusters. Then for each cluster, we normalized the scores of keywords and selected the top 3. The result is organized in table 4.

Furthermore, the clustering makes sense because similar movies with similar keywords are being grouped together using NLP techniques. For example, movies in cluster 1 all focus on the themes of war, conflict, and survival and have excellent storytelling.

Table 4

	Movies	Keywords
Cluster 1	Schindler's List, The Matrix, Life is Beautiful, The Pianist, The Usual Suspects	Story, war, family
Cluster 2	BTS World Tour Love Yourself in Seoul, David Attenborough: A Life on Our Planet, Earthlings	World, life, man
Cluster 3	The Godfather, The Godfather Part II, Persona, Harakiri, Spirited Away, The Best of Youth, Whiplash	Family, young, crime
Cluster 4	The Shawshank Redemption, The Lord of the Rings: The Return of the King, The Lord of the Rings: The Fellowship of the Ring, Pulp Fiction, The Lord of the Rings: The Two Towers, Inception, Fight Club, …	Life, man, American
Cluster 5	The Dark Knight, 12 Angry Men, One Flew Over the Cuckoo's Nest, Dear Zachary: A Letter to a Son About His father, The Departed	Soon, American, young

Conclusion

In this study, we explored the variables contributing to movie success through a robust data-driven approach. The raw data were sourced from IMDb, TMDb, and Rotten Tomatoes webpages and preprocessed. We investigated the relationships between these different ratings, movie budget trends, directors and production company success factors, and movie group keywords. More specifically, we gained understanding of the trends and patterns in the film industry and demonstrated a moderate to minor level of correlation between rating platforms, suggesting that each has its nuances in movie evaluation. In addition, the visualization of budget trends showed an overall increase over the years, and for directors and production companies, our results showed the important impact of high-grossing movies in determining their rankings. Then with the use of TF-IDF and clustering methods, we produced the major thematic groupings in movie descriptions generated by combinations of keywords. The resulting clusters with keywords ranging from war to family to American life covered a diverse collection of top rated movies.

Lastly, it should be noted that our study could be further improved. For example, the dataset could be expanded to include more movies for more accurate data insights. Challenges encountered in data preprocessing underscore areas for further improvement and work. That being said, our findings offered a meaningful framework for understanding the dynamics and patterns of the film industry and paved the way for more systematic data exploration.